

Event-State Theory VI: General Relativity as the Infrared Effective Theory of Discrete Causal Optimization

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We present a constrained derivation sketch showing how the Einstein field equations arise as the infrared effective description of a discrete successor-state selection principle. Event-State Theory (EST) posits that the universe evolves through a sequence of discrete global states $\{\Psi_n\}$, where admissible successor states are selected by minimizing (or probabilistically weighting) a computational cost functional $J(\Psi_n \rightarrow \Psi_{n+1})$. Focusing on the geometric contribution J_{geom} —the cost associated with the causal/update structure itself—we argue that systematic anisotropies in local update neighborhoods constitute an intrinsic “lattice tax”. Under coarse-graining, the macroscopic response to reducing this tax is encoded by an emergent metric field. Assuming locality, emergent diffeomorphism invariance, and a second-order infrared truncation, the leading continuum action is constrained to the Einstein–Hilbert form (plus a cosmological constant and higher-derivative corrections). The Einstein equations follow by variation. We explicitly distinguish (I) assumptions required for the infrared effective field theory step from (II) open problems needed for a full microscopic-to-continuum coefficient match.

I. INTRODUCTION

General Relativity (GR) describes gravity as spacetime curvature, but the standard formulation does not specify a microscopic origin for why geometry should be dynamical. Event-State Theory (EST) proposes a discrete, pre-geometric substrate in which the universe is described by a sequence of global event-states $\{\Psi_n\}$. Successor states are selected according to a cost functional J (deterministically or probabilistically), implementing the Path of Causal Least Resistance (PCLR).

This paper addresses the *geometric* sector of the theory: how the causal/update structure itself is selected. The core idea is that discrete update neighborhoods generically exhibit anisotropic directionality. If anisotropy induces additional computational overhead, then reducing that overhead produces an effective macroscopic description equivalent to GR.

A. EST framework (minimal)

EST assumes:

$$\Psi_0, \Psi_1, \Psi_2, \dots \quad (1)$$

$$\Psi_{n+1} \in \Omega(\Psi_n), \quad (2)$$

$$\Psi_{n+1} = \arg \min_{\phi \in \Omega(\Psi_n)} J(\Psi_n \rightarrow \phi) \quad \text{or} \quad P(\phi|\Psi_n) \propto e^{-J/\lambda}. \quad (3)$$

The total cost decomposes schematically as

$$J_{\text{total}} = J_{\text{geom}}[G] + J_{\text{matter}}[\text{fields on } G] + \dots \quad (4)$$

This paper focuses on J_{geom} .

II. DISCRETE CAUSAL SUBSTRATE AND UPDATE ANISOTROPY

A. Causal graph per step

Represent each state by a directed graph

$$G_n = (V_n, E_n), \quad (5)$$

where vertices are events and edges encode update/causal dependence.

B. Local update stencils

For $v \in V_n$, define the update neighborhood

$$\mathcal{S}_n(v) = \{v' \in V_n : v' \text{ influences the update of } v\}. \quad (6)$$

In a regular lattice $\mathcal{S}$ is isotropic; in a generic discrete substrate it can be irregular and directionally biased.

C. Anisotropy tensor

Assign to each neighbor $v' \in \mathcal{S}(v)$ a direction unit vector

$$\hat{n}(v \rightarrow v') = \frac{\mathbf{x}(v') - \mathbf{x}(v)}{\|\mathbf{x}(v') - \mathbf{x}(v)\|}. \quad (7)$$

Define the (symmetric, traceless) anisotropy tensor

$$\Delta^{ab}(v) = \frac{1}{|\mathcal{S}(v)|} \sum_{v' \in \mathcal{S}(v)} \left[n^a n^b - \frac{1}{3} \delta^{ab} \right]. \quad (8)$$

$\Delta^{ab} = 0$ iff the stencil is isotropic.

D. Assumption: anisotropy costs

Assumption A1 (lattice tax): anisotropic update neighborhoods impose additional computational overhead (relative to isotropic neighborhoods), therefore J_{geom} penalizes $\Delta \neq 0$.

III. GEOMETRIC COST FUNCTIONAL: LOCALITY AND SYMMETRY

A. Local derivative expansion

Assumption A2 (IR locality): after coarse-graining, the geometric cost admits a local effective description:

$$J_{\text{geom}} \rightarrow S_{\text{geom}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{geom}}. \quad (9)$$

B. Leading scalar from anisotropy

Rotational symmetry implies the leading contribution is quadratic:

$$\mathcal{L}_{\Delta} = c_0 + c_2 \Delta_{ab} \Delta^{ab} + O(\Delta^3). \quad (10)$$

IV. COARSE-GRAINING AND THE EMERGENT METRIC

Partition the microscopic graph into blocks B_i with many events and define the block-averaged anisotropy

$$\bar{\Delta}^{ab}(B_i) = \frac{1}{|B_i|} \sum_{v \in B_i} \Delta^{ab}(v). \quad (11)$$

In the continuum limit, $\bar{\Delta}^{ab} \rightarrow \Delta^{ab}(x)$ becomes a smooth field.

A. Metric identification (controlled ansatz)

Assumption A3 (metric encoding): the cheapest macroscopic representation of persistent anisotropy is geometric deformation, encoded by an emergent metric perturbation:

$$g_{ab} = \eta_{ab} + \epsilon h_{ab}, \quad h_{ab} \sim \Delta_{ab}. \quad (12)$$

A complete microscopic derivation requires an explicit homogenization map $\mathcal{H} : \Delta \mapsto h$; here we use (12) as the controlled identification required for the IR effective theory step.

V. EMERGENCE OF TIME AND LORENTZIAN SIGNATURE

EST posits no fundamental time coordinate; only a discrete ordering of successor states.

A. Two relations: co-presence and precedence

Events exhibit: (i) *co-presence* (same Ψ_n , interpreted as spatial relation), (ii) *precedence* ($v \prec v'$, interpreted as causal/update ordering).

B. Finite propagation and causal cones

Because updates depend on a finite stencil range, influence from an event v cannot reach arbitrary events in one step. After k steps, the reachable set grows outward, defining a discrete causal “cone” structure. In the continuum limit this yields a partition into timelike (reachable), spacelike (not reachable), and null (boundary) separations.

C. Lorentzian signature as the continuum encoding

A continuum metric must encode this trichotomy; therefore the effective macroscopic metric has Lorentzian signature $(-, +, +, +)$, with the timelike direction corresponding to successor ordering. No fundamental t is assumed; the time parameter is an emergent macroscopic label for causal progression.

VI. WHY THE EINSTEIN–HILBERT TERM APPEARS IN THE IR

A. Emergent diffeomorphism invariance

Assumption A4: coarse-graining renders the macroscopic description insensitive to microscopic labeling choices, which becomes diffeomorphism invariance in the continuum.

B. Second-order infrared truncation

Assumption A5: the dominant IR field equations are second order, corresponding to an effective Lagrangian built from $g_{\mu\nu}$ and at most two derivatives.

C. Uniqueness argument (Lovelock/EFT proof sketch)

Given locality, diffeomorphism invariance, and at most two derivatives of the metric, the leading scalar density

in four dimensions is constrained to

$$\sqrt{-g}(a_0 + a_1 R), \quad (13)$$

up to a total derivative, with higher-derivative invariants (e.g. R^2 , $R_{\mu\nu}R^{\mu\nu}$) suppressed by the coarse-graining scale. This is the standard effective-field-theory logic often summarized by Lovelock-type uniqueness results for second-order metric field equations.

Therefore, the macroscopic effective action takes the form

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \left(\Lambda + \frac{1}{16\pi G_{\text{eff}}} R \right) + S_{\text{matter}} + \dots \quad (14)$$

VII. EINSTEIN FIELD EQUATIONS

Varying (14) with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}. \quad (15)$$

Energy–momentum conservation follows from diffeomorphism invariance via the Bianchi identity.

VIII. WHAT REMAINS OPEN (FOR A FULL DERIVATION)

This paper constrains the IR form of the effective theory under explicit assumptions. Completing the microscopic-to-continuum derivation requires:

1. A precise microscopic definition of J_{geom} in terms of update neighborhoods/stencil statistics.
2. An explicit homogenization map $\mathcal{H} : \Delta^{ab} \mapsto h_{\mu\nu}$ with controlled approximations.
3. Coefficient matching: computing or bounding G_{eff} and Λ from microscopic parameters (analytically or numerically).

IX. CONCLUSION

Under EST’s discrete successor-selection principle, anisotropic update neighborhoods naturally motivate a geometric cost (a lattice tax). If the macroscopic system reduces this tax by encoding it geometrically, then coarse-graining yields an emergent Lorentzian metric. Assuming locality, emergent diffeomorphism invariance, and a second-order infrared truncation, the leading effective action is constrained to the Einstein–Hilbert form (plus cosmological constant and higher-derivative corrections). The Einstein field equations follow by variation. This establishes GR as the infrared effective theory of a discrete cost-optimizing substrate, while clearly separating the EFT-constrained result from the remaining microscopic coefficient-matching program.

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